Evidence at What Cost? Decoupling and Scaling in LLM-Based Document Relation Extraction

Anonymous ACL submission

Abstract

LLM-based Document-level Relation Extraction (DocRE) now matches or exceeds encoder baselines in extraction accuracy, yet these methods output bare relation triplets without identifying supporting sentences. Encoder-based evidence methods (SAIS, DREEAM) provide supporting sentences but trail LLM approaches in accuracy, creating a gap between extraction performance and output verifiability. We formalize evidence-grounded DocRE as structured generation and compare GRPO, DPO, and RSFT across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. The evidence tax—extraction cost of generating evidence—is scale-dependent: beneficial at 1.7B (-5.63pp dev Ign-F1) but approximately neutral at $\geq$4B (+0.25$\pm$0.47pp at 4B; positive = cost). Gradient-based optimization amplifies evidence-extraction competition (GRPO catastrophic, DPO gradual), while RSFT with NLI-based scoring avoids decoupling entirely through output filtering rather than gradient pressure. RSFT consistently reduces training seed variance (complete pairwise seed separation on LLaMA, Cohen's d=2.58; non-overlapping χ^2 CIs at 1.7B). Our 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B SFT performance (50.19, std 3.23pp) at 4.7$\times$ fewer parameters. Test-set evaluation preserves model ranking across all evaluated scales (Spearman ρ=1.0).

1 Introduction

Document-level relation extraction (DocRE) requires identifying semantic relations between entity pairs across multiple sentences, a task where encoder-based methods such as Zhou et al. (2021) long dominated. Recent LLM-based approaches have closed this gap: Xue et al. (2024) fine-tune LLMs with structured decomposition, Li et al. (2024) frame it as multi-choice reading comprehension, and Xu et al. (2025) combine retrieval-augmented generation with structured knowledge. These methods approach or surpass encoder baselines on DocRED (Yao et al., 2019) and Re-DocRED (Tan et al., 2022) in extraction accuracy, yet every one outputs bare relation triplets (e_h, r, e_t) without indicating *which sentences* support each prediction. In deployment contexts—clinical knowledge bases, legal due-diligence pipelines, financial compliance systems—users cannot verify a prediction without manually locating its textual basis across a multi-page document.

Encoder-based methods have partially addressed this need: Xiao et al. (2022) introduce sentence-level evidence supervision as an auxiliary signal, and Ma et al. (2023) extend this with dynamic evidence retrieval through memory attention. Both produce evidence and improve extraction, but encoder methods now trail LLM-based approaches in accuracy. The community faces a gap: the most accurate extractors provide no evidence, while evidence-capable methods are no longer competitive.

This gap reflects a structural difference. In encoder methods, evidence retrieval is classification that regularizes the encoder at negligible cost; in LLM-based DocRE, evidence requires the model to *generate* additional output tokens per triple, increasing output length and decoding complexity. The naïve expectation is that this generation cost always harms extraction accuracy. Our experiments show this expectation is wrong. The evidence tax is not a fixed penalty but a landscape shaped by model capacity, optimization paradigm, and training data quality. Within the Qwen3 family, evidence generation *benefits* extraction at 1.7B (-5.63pp Ign-F1, partly through halving output duplication) but is approximately neutral at $\geq$4B (+0.25$\pm$0.47pp at 4B). Gradient-based optimization amplifies this competition—GRPO collapses extraction under constrained precision, DPO pro-

gressively trades extraction for evidence—while discrete sample selection (RSFT) avoids the trade-off entirely by filtering outputs rather than pushing gradients.

We formalize evidence-grounded DocRE as a structured generation task and compare three post-training paradigms—GRPO (Shao et al., 2024), DPO (Rafailov et al., 2023), and RSFT (building on STaR; Zelikman et al. 2022)—across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. Our contributions:

1. **Evidence generation is not uniformly harmful.** We observe a scale-dependent evidence tax with a within-family capacity threshold: beneficial at 1.7B (-5.63pp) but approximately neutral at $\geq$4B ($+0.25 \pm 0.47$pp at 4B), challenging the assumption that evidence always costs extraction accuracy.

2. **Gradient pressure causes evidence-extraction decoupling.** We characterize a decoupling spectrum (GRPO catastrophic $\rightarrow$ DPO gradual $\rightarrow$ RSFT none) and introduce the Evidence-Decoupling Ratio (EDCR) as a diagnostic, validated at per-relation level ($\rho = -0.79$, $n = 76$, $p < 0.001$).

3. **RSFT consistently reduces training variance.** NLI-based sample selection improves mean Ign-F1 across tested scales (+1.2 to +6.6pp); our 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B SFT performance (50.19, std 3.23pp) at 4.7$\times$ fewer parameters, enabling deployment-efficient evidence-grounded extraction.

2 Related Work

Document-level relation extraction (DocRE) requires reasoning across sentence boundaries to identify relations between entity pairs. DocRED (Yao et al., 2019) established the first large-scale benchmark for this task, catalyzing a line of encoder-based methods that treat DocRE as classification over entity pair representations. ATLOP (Zhou et al., 2021) introduced adaptive thresholding for multi-label prediction and localized context pooling for entity representations. A parallel thread incorporates sentence-level evidence into these architectures. SAIS (Xiao et al., 2022) introduced evidence supervision as an auxiliary training signal and showed that joint evidence prediction improves extraction through data augmentation and ensemble inference. DREEAM (Ma et al., 2023) extended this approach with dynamic memory-based evidence retrieval that guides attention to supporting sentences. EIDER (Xie et al., 2022) jointly trains an evidence extractor and fuses evidence representations into the relation classifier. In all three systems, evidence identification is binary sentence classification over a fixed candidate set, a formulation that adds no output complexity. Recent work has recast DocRE for large language models. AutoRE (Xue et al., 2024) fine-tunes LLMs with QLoRA using a relation-head-facts decomposition; Li et al. (2024) frame extraction as multi-choice reading comprehension; RelPrior (Pi et al., 2025) uses relation priors to guide LLM-based extraction; GLiDRE (Armingaud and Besançon, 2025) presents a generalist lightweight approach; Ru-LLM (Zhang et al., 2025) incorporates logical rule constraints; and Xu et al. (2025) combine retrieval-augmented generation with structured knowledge. These methods match or surpass encoder baselines but produce bare triplets without evidence. Among LLM-based generative methods at the 8B scale, our RSFT model (54.33 Ign-F1 on DocRED dev) outperforms AutoRE (52.14[1]) and is slightly below RelPrior (-1.17pp), while being the only method that additionally provides sentence-level evidence grounding. The critical distinction lies in computational nature: in SAIS and DREEAM, evidence retrieval is classification that regularizes the encoder without increasing output length; in LLM-based DocRE, evidence requires additional output tokens per triple, imposing costs that depend on model capacity and optimization strategy. Whether this cost is inevitable or can be mitigated is the central question of our work.

Group Relative Policy Optimization (GRPO; Shao et al. 2024) eliminates the separate value network by computing advantages within groups. Subsequent work addresses failure modes: DAPO (Yu et al., 2025) prevents entropy collapse, Dr. GRPO (Liu et al., 2025) corrects length bias, and multi-objective variants (Liu et al., 2026; Ichihara et al., 2025) handle competing reward signals. Applications to structured IE include R1-RE (Dai et al., 2025) and REAR (Luo et al., 2025). Direct Preference Optimization (Rafailov et al., 2023) reformulates RLHF as implicit reward maximization. Our

[1]AutoRE Ign-F1 is from the LLaMA3-8B reproduction by Pi et al. (2025); the original paper reports only F1.

decoupling spectrum is distinct from catastrophic forgetting in RL alignment (Jin et al., 2025), where models lose pre-trained capabilities across tasks; in our setting, extraction and evidence compete *within* a single task, resembling negative transfer in multi-task learning (Ruder, 2017; Standley et al., 2020; Yu et al., 2020), amplified by QLoRA's constrained parameter space (Gao et al., 2023; Moskovitz et al., 2024). RSFT avoids this intra-task competition by operating on discrete outputs rather than gradients, extending the sample-and-select principle (Wang et al., 2023) to structured IE with joint quality scoring.

Evidence faithfulness evaluation has received increasing attention: Laban et al. (2022) re-visit NLI-based models for summarization inconsistency detection, Li et al. (2025a) propose dual-reward inference for faithful rationale generation, Li et al. (2025b) use supervised rationale verification with contrastive feedback for relation extraction, and Zhang et al. (2024) introduce fine-grained NLI-based faithfulness metrics for summarization. Our CED scoring builds on similar NLI-based principles but targets evidence-relation entailment rather than free-form rationales.

Zelikman et al. (2022) introduced STaR for iterative self-improvement; Dong et al. (2023) generalize this with reward-model-based sample filtering; recent analysis shows rejection sampling implicitly performs PPO-like optimization (Anil et al., 2025). Our RSFT is a single-round variant applied to evidence-grounded DocRE with CED/NLI scoring as a domain-specific quality filter. Because RSFT selects complete outputs rather than differentiating through a reward signal, it cannot exploit the extraction-evidence trade-off that gradient methods discover, inherently preserving joint quality.

3 Method

3.1 Task: Evidence-Grounded Document-Level Relation Extraction

Given a document $D = \{s_1, \ldots, s_n\}$ consisting of n sentences and a set of entities $\mathcal{E}$ identified within D, the task of evidence-grounded document-level relation extraction (DocRE) requires predicting, for each entity pair $(e_h, e_t) \in \mathcal{E} \times \mathcal{E}$, a relation $r \in \mathcal{R} \cup \{\texttt{NA}\}$ together with a set of supporting evidence sentences $E \subseteq D$. Standard DocRE predicts only the triple (e_h, r, e_t); the evidence-grounded formulation also requires E, which grounds each prediction in specific textual support (Figure 1).

In our LLM-based approach, the model generates structured outputs that jointly encode both components. For each predicted relation, the output follows the format `(head, relation, tail) | Evidence: [S`i`][S`j`]`, requiring the model to produce complete evidence references inline with each triple rather than as a separate retrieval step. This design contrasts with encoder-based evidence methods such as SAIS (Xiao et al., 2022) and DREEAM (Ma et al., 2023), which frame evidence identification as sentence-level binary classification. Because classification over a fixed sentence set does not increase output length, these methods incur no generation overhead. Our LLM formulation requires additional tokens for each evidence reference, creating a generation burden that scales with the number of predicted triples and their evidence sentences. This scale-dependent cost is the central phenomenon we investigate.

3.2 Supervised Fine-Tuning Baseline

We fine-tune all models with LoRA (Hu et al., 2022) at rank r=64 and scaling factor α=128, applied to all linear projections (attention and feed-forward). The training data consists of DocRED (Yao et al., 2019) training documents converted to the structured output format described above, where each example maps a document and its entity set to the complete set of relation triples with evidence annotations. We also train a *no-evidence* variant under identical configuration but with evidence references removed from the output format, producing only `(head, relation, tail)`. This variant serves as an upper bound on extraction performance by eliminating the generation cost of evidence, and allows direct measurement of the evidence tax at each model scale. Detailed hyperparameters (learning rate, batch size, number of epochs) are reported in the Appendix.

3.3 Optimization Paradigms

We evaluate three post-SFT optimization paradigms that differ in how they incorporate quality signals from the CED scoring function (§3.4).

GRPO. Group Relative Policy Optimization (Shao et al., 2024) is an online RL method that eliminates the need for a separate value network by computing advantages within groups of sampled outputs. For each training document, the policy generates G candidate responses, which are scored

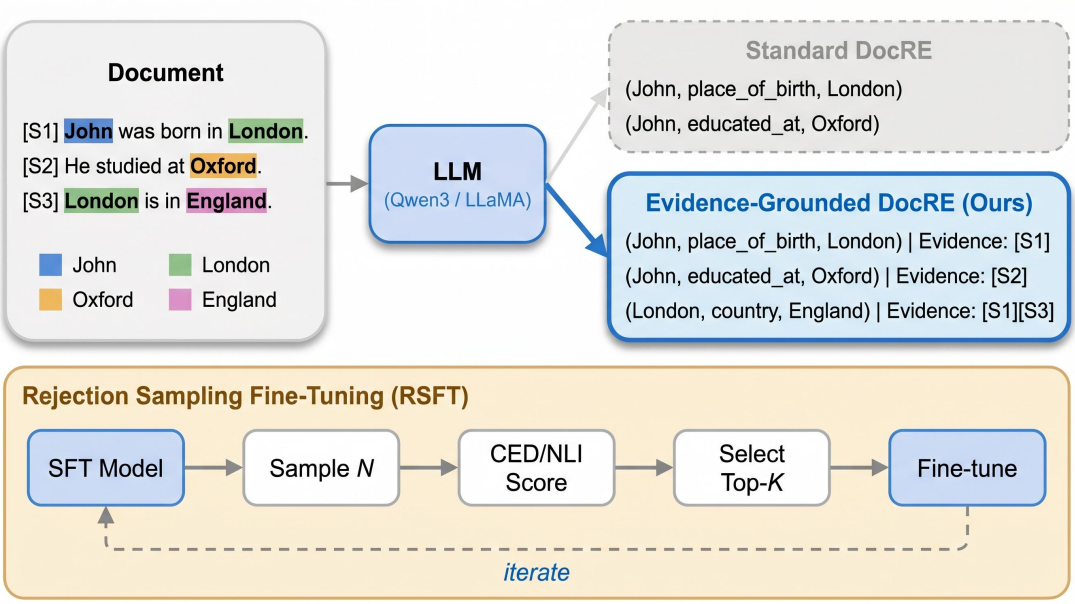


Figure 1: Overview of our approach. (Top) Standard DocRE outputs bare triples; evidence-grounded DocRE additionally generates supporting sentence indices for each prediction. (Bottom) RSFT pipeline: the SFT model samples N candidate responses per document, scores them with CED/NLI, selects the top-k, and fine-tunes on the filtered set.

by CED. The advantage of each response is computed relative to the group mean, and the policy is updated to increase the probability of above-average outputs. We test two configurations: QLoRA with G=4 groups under 4-bit quantization, and bf16 full-precision LoRA with G=8. Because GRPO applies gradient updates that directly optimize the reward signal, evidence quality and extraction accuracy compete for gradient capacity. We analyze this multi-objective tension in Section 4.3.

DPO. Direct Preference Optimization (Rafailov et al., 2023) reformulates RLHF as a classification problem over preference pairs, without explicit reward models. We construct preference data from the SFT model's sampled outputs: for each document, the response with the highest CED score is labeled as preferred and the lowest-scoring response as rejected. DPO then adjusts the policy to increase the log-probability ratio between preferred and rejected completions. We train for 1 and 3 epochs to observe the temporal dynamics of evidence-extraction trade-offs under implicit optimization.

RSFT. Rejection Sampling Fine-Tuning is a single-round variant of iterative self-improvement methods such as STaR (Zelikman et al., 2022). The procedure is straightforward: the SFT model generates N=8 candidate outputs per training document, each candidate is scored by CED, and only the top-k=1 response is retained as training data. We use k=1 (best-of-N) selection throughout all experiments, selecting the single highest-scoring candidate per document for RSFT training data construction. This yields approximately 3,000 training examples per model (one per document; exact counts vary slightly by model due to generation failures). Unlike GRPO and DPO, RSFT does not apply gradient-based optimization pressure toward a reward signal. Instead, it improves training data quality through filtering: the model trains on its own best outputs rather than all outputs. This distinction is critical because discrete selection over complete outputs cannot exploit the extraction-evidence trade-off that gradient methods find. The contribution of RSFT in our setting is not the framework itself, which is well-established, but its application to evidence-grounded DocRE with CED as a domain-specific quality filter.

3.4 Contrastive Evidence Discrimination

CED is a scoring function that evaluates the faithfulness of predicted evidence sentences to their associated relation triples. It builds on Natural Language Inference (NLI) to measure whether evidence actually entails the predicted relation, rather than merely co-occurring with the relevant entities.

For a predicted triple (e_h, r, e_t) with evidence sentences E, CED constructs a positive NLI pair: the premise is the concatenation of sentences in E, and the hypothesis is a natural-language verbalization of the relation (e.g., "e_h has relation r with e_t"). An NLI model (`cross-encoder/nli-deberta-v3-base`) produces an entailment probability p^+ for this pair. The contrastive component introduces hard negatives drawn from the same document. For each triple, we collect sentences that mention e_h or e_t but are not in the predicted evidence set. These entity-sharing distractors serve as negative premises, and the NLI model scores each to obtain $p_1^-, \ldots, p_m^-$. The CED score for a single triple is:

$$\text{CED}(e_h, r, e_t, E) = p^+ - \max_j p_j^- \qquad (1)$$

This formulation forces the scoring function to distinguish sentences that genuinely support the relation from sentences that merely mention the same entities. A high CED score indicates that the evidence provides relation-specific support beyond entity co-occurrence.

The document-level CED score aggregates over all predicted triples, weighted by extraction correctness: triples that match ground-truth relations contribute their CED scores, while incorrect triples receive zero credit. This coupling ensures that CED rewards both faithful evidence and accurate extraction jointly.

We also evaluate a *flat-NLI* baseline that uses only the positive entailment score p^+ without the contrastive term. CED and flat-NLI produce comparable RSFT outcomes (mean Rel-F1 difference of 0.02pp across available seeds; Section 4.7), confirming that the contrastive formulation is not essential for data selection quality. We adopt CED as the default because its per-triple discrimination scores support interpretable error analysis.

3.5 Evidence-Decoupling Ratio

We introduce the Evidence-Decoupling Ratio (EDCR) as a diagnostic metric that quantifies the degree of misalignment between a model's extraction and evidence capabilities. EDCR is not a training component; it serves purely as an analytical tool for comparing optimization methods.

EDCR is defined as the distractor citation rate across all predicted triples:

$$\text{EDCR} = \frac{\sum_i |E_i^{\text{pred}} \setminus E_i^{\text{gold}}|}{\sum_i |E_i^{\text{pred}}|} \qquad (2)$$

where the sum ranges over all predicted triples with at least one evidence citation. E_i^{pred} is the set of evidence sentences cited for predicted triple i, and E_i^{gold} is the corresponding gold evidence (empty for false-positive predictions). EDCR thus measures the fraction of all cited evidence that does not appear in the gold evidence for the associated triple.

A model with many false-positive triples will have high EDCR because all evidence for incorrect triples counts as distractor; even among correct triples, citing irrelevant sentences inflates the numerator. Conversely, accurate extraction with faithful evidence yields low EDCR. This dual sensitivity makes EDCR a natural diagnostic for extraction-evidence decoupling.

EDCR operates at two diagnostic levels. At the method level, comparing EDCR across seven independent method configurations reveals a negative correlation with Rel-F1 (Spearman $\rho{=}-0.89$, $n{=}7$, $p{=}0.007$). Because EDCR is mechanically influenced by false-positive triples (which share determinants with Rel-F1), this correlation should be interpreted as a diagnostic pattern rather than an independent discovery; its primary value lies in per-relation analysis. We validate EDCR's diagnostic utility through per-relation analysis (§4.7), which avoids the shared-component concern present in cross-method correlation. At the per-relation level, computing EDCR for individual relation types reveals a robust negative correlation with Rel-F1 ($\rho{=}-0.79$, $n{=}76$, $p{<}0.001$) that is independent of Evi-F1 ($\rho{=}-0.06$, $p{=}0.58$). This independence means EDCR captures relation-specific extraction-evidence misalignment that aggregate evidence metrics miss. The SFT baseline EDCR (0.6797, three-seed mean) reflects the inherent distractor rate under standard training; EDCR values substantially exceeding this level (e.g., GRPO QLoRA at 0.9876) indicate RL-induced decoupling beyond what evidence production alone explains, while values below the baseline (e.g., RSFT at 0.6487) reflect improved co-optimization.

4 Experiments

4.1 Experimental Setup

We evaluate on DocRED (Yao et al., 2019) (3,053 training / 998 dev documents, 96 relation types) with Re-DocRED (Tan et al., 2022) for cross-dataset validation. Our primary model is Qwen3-4B; we also train Qwen3-1.7B, 8B, and 14B to

Table 1: Main results on DocRED dev (Qwen3-4B). Ign-F1 shows mean ± std$^{(\text{seeds})}$; other metrics are same-seed means. Best **bold**; best evidence-generating underlined. Per-seed: Appendix 4.

Method	Ign-F1	Rel-F1	Evi-F1	EDCR
SFT baseline	43.58±3.80$^{(3)}$	44.39	79.61	0.6797
SFT no-evidence	**47.21**	**48.76**	–	–
RSFT CED	44.82±0.51$^{(3)}$	45.80	80.96	0.6487

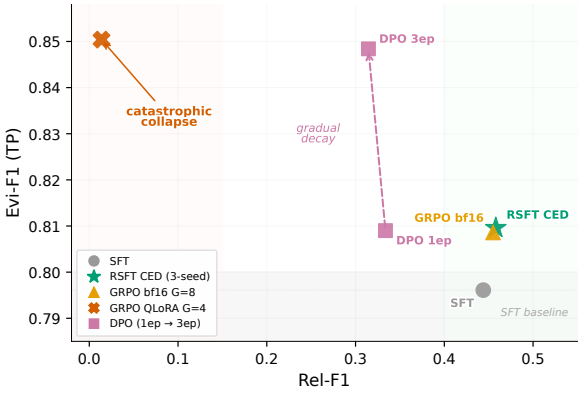


Figure 2: Decoupling spectrum. Each method is plotted by Rel-F1 (x-axis) and Evi-F1 (y-axis). Arrows indicate training progression for DPO and GRPO bf16. Decoupled methods drift toward the upper-left.

study scale effects, with LLaMA-3.1-8B for cross-family validation. All models use LoRA (Hu et al., 2022) (r=64, α=128) with structured output jointly producing triples and evidence. RSFT selects the top-k=1 of N=8 samples by CED score (§3.4); GRPO uses group sizes G=4 (QLoRA) and G=8 (bf16); DPO constructs preference pairs from CED-ranked outputs. We report Ign-F1 (primary; ignores training-set triples), Rel-F1, Evi-F1 (TP-only and joint per DREEAM protocol (Ma et al., 2023)), and EDCR (§3.5). Detailed hyperparameters appear in the Appendix.

4.2 Main Results

Table 1 presents the primary comparison on Qwen3-4B. The SFT baseline exhibits high seed sensitivity (mean Ign-F1 43.58, std 3.80pp, range 39.74–47.34). Removing evidence entirely (no-evidence SFT, s42) reaches 47.21 Ign-F1.

RSFT CED achieves 44.82 Ign-F1 (+1.24 over SFT mean) with substantially lower variance (std 0.51pp), though these reflect different sources of variation (Section 4.4). Even the weakest RSFT seed (44.23) exceeds the SFT mean. Evidence quality improves (Evi-F1 80.96 vs. 79.61) and coupling tightens (EDCR 0.6487 vs. 0.6797). Per-seed results appear in Appendix B.

4.3 Evidence-Extraction Decoupling Spectrum

Because evidence and extraction share gradient capacity, different optimization paradigms handle this competition in qualitatively different ways. We organize the outcomes along a *decoupling spectrum* from catastrophic collapse to no decoupling (Figure 2; full data in Table 13, Appendix; training dynamics in Figure 5, Appendix).

GRPO. Under QLoRA 4-bit quantization (G=4), GRPO drives Rel-F1 to 1.38 while Evi-F1 rises to 85.04 (EDCR 0.9876): quantization-constrained precision limits the effective parameter space, so the reward signal preferentially optimizes the easier sub-task (Gao et al., 2023), resembling the quantized-actor divergence of Li et al. (2026). Full-precision bf16 (G=8) avoids catastrophe (Rel-F1 45.49, EDCR 0.6318), confirming quantization precision rather than group size as the primary factor.

DPO and RSFT. DPO exhibits substantial extraction degradation: Rel-F1 drops from 33.37 (1 epoch) to 31.45 (3 epochs) while Evi-F1 rises (80.90→84.84). EDCR is non-monotonic (0.8123→0.7503): despite worsening extraction, evidence selectivity improves with continued training, partially reducing the decoupling. RSFT maintains both capabilities simultaneously (Rel-F1 45.80, Evi-F1 80.96, EDCR 0.6487) because discrete output selection cannot exploit the extraction-evidence trade-off.

4.4 Scale-Dependent Evidence Tax

The evidence tax, defined as Ign-F1(no-evidence) − Ign-F1(with-evidence) on the development set, is scale-dependent. Under matched evaluation conditions (`max_new_tokens`=2048, format-compliant intersection; Appendix F), the training-time tax reveals a within-family capacity threshold: evidence generation benefits 1.7B (−5.63pp; Appendix F), where it also halves the output duplication rate (60.5%→33.0%; Table 7 in Appendix E), but is approximately neutral at ≥4B (+0.25±0.47pp at 4B, n=3; +0.04pp at 8B; +0.03pp at 14B). Table 2 and Figure 3 present RSFT performance and inference-time evidence tax across five models (Figure 4 in Appendix). Evidence training also affects SFT seed variance in a scale-dependent manner: at 4B and 8B, with-evidence models show lower cross-

seed spread than no-evidence models (25× and 1.4× compression, respectively), but at 1.7B the pattern reverses—with-evidence variance exceeds no-evidence variance, likely amplified by extreme truncation rates at this scale (Appendix F).

RSFT navigates this capacity landscape through quality-based data selection rather than gradient pressure. Mean Ign-F1 improves across tested scales: +6.55pp at 1.7B, +1.24pp at 4B, +4.14pp at 8B, and +3.44pp at 14B (Table 2). RSFT from an independent SFT base (s44, Method B) achieves 51.52 Ign-F1 at 4B (+4.18pp over that base). This result confirms that RSFT gains are not contingent on a specific SFT checkpoint.

Beyond mean performance, RSFT consistently reduces training variance (sign test: 4/4 scales, p=0.0625; χ^2 CI analysis confirms non-overlapping intervals at the 1.7B scale; Cohen's $d \geq 1.79$ for 3/4 scales; Appendix D). Three RSFT seeds from a shared SFT checkpoint yield standard deviations of 0.09pp at 1.7B, 0.51pp at 4B, and 0.50pp at 8B (n=3 each), compared to SFT cross-seed variation of 2.79pp, 3.80pp, and 3.23pp respectively. While these measure different sources of variance (SFT reflects full training runs from different random seeds; RSFT reflects additional fine-tuning from a fixed base), the practical consequence is direct: across all Qwen3 scales with ≥3 seeds, variance compression is 32.6× at 1.7B, 7.5× at 4B, and 6.5× at 8B (point estimates from N=3 seeds; confidence intervals are wide, Appendix D). Per-scale Rel-F1, Evi-F1, and EDCR breakdowns appear in Table 5 (Appendix C). The 1.7B RSFT model (51.19 Ign-F1, std 0.09pp) approaches 8B SFT performance (50.19, std 3.23pp) at 4.7× fewer parameters—comparable in mean yet 36× more predictable.

Cross-family validation on LLaMA-3.1-8B confirms these patterns. All three RSFT seeds (55.87–57.14) exceed all three SFT seeds (48.02–53.84), yielding complete pairwise seed separation (Cohen's d=2.58) and 4.6× variance compression (std 2.92pp → 0.64pp). The SFT seed lottery is starkest at s44 (48.02 Ign-F1, 24.4% truncation, elevated Evi-F1 85%), a degenerate outcome that RSFT eliminates entirely (s44: 55.87). This robustness extends to inference-time format changes: LLaMA SFT models evaluated without evidence at 1024 tokens suffer 38–42% truncation with Ign-F1 collapsing to ∼0.33 (s43/s44), while RSFT models truncate only 18–19% with stable performance (55.41±0.13pp), indicating that CED filtering selects for compact output structure.

Table 2: RSFT performance and inference-time evidence tax across model scales (DocRED dev Ign-F1). LLaMA-3.1-8B included for cross-family validation. Mean ± std$^{(\text{seeds})}$.

Model	SFT w/ evi	RSFT	RSFT Tax†	RSFT Gain
Qwen3-1.7B	44.64±2.79$^{(3)}$	51.19±0.09$^{(3)}$	−7.75	+6.55
Qwen3-4B	43.58±3.80$^{(3)}$	44.82±0.51$^{(3)}$	+0.94	+1.24
Qwen3-8B	50.19±3.23$^{(3)}$	54.33±0.50$^{(3)}$	−0.78	+4.14
Qwen3-14B	55.07$^{(1)}$	58.51±0.50$^{(2)}$	−2.78±2.49$^{(2)}$	+3.44
LLaMA-3.1-8B	51.09±2.92$^{(3)}$	56.54±0.64$^{(3)}$	−2.17±0.76$^{(3)}$	+5.45

†Inference-time: same RSFT model with vs. without evidence output. No-evi eval uses `max_new_tokens`=2048; with-evi eval uses 1024 (1.7B uses 2048 for both). Tax = no-evi − with-evi; positive = cost. Seeds: 1.7B s42; 4B Method B s44; 8B s42; 14B two-seed mean; LLaMA three-seed mean. Training-time tax (matched@2048 fair evaluation) in Appendix F.

For RSFT models, the inference-time evidence tax (same model evaluated with vs. without evidence output) preserves the benefit direction at 1.7B (−7.75pp, confirmed with 2048-token no-evidence evaluation that rules out truncation). At 4B, where the training-time tax is approximately neutral (+0.25±0.47pp; Appendix F), the RSFT inference-time tax is similarly small (+0.94pp). At 8B, the tax reverses to a benefit (−0.78pp), and at 14B a benefit is observed (−2.78±2.49pp, two-seed mean), directionally consistent with the 8B pattern though the CI crosses zero at n=2. RSFT thus maintains or improves extraction quality with evidence across the Qwen3 family. LLaMA-3.1-8B RSFT shows a larger inference-time evidence benefit (−2.17±0.76pp, three-seed mean, vs. −0.78pp for Qwen3-8B), confirming that both performance gains and training reliability generalize across model families.

The effect of RSFT on inference-time evidence dependence is heterogeneous. At 1.7B, RSFT maintains strong evidence benefit (−5.01±0.59pp, two-seed mean), comparable to SFT (−4.86pp). At 4B, RSFT introduces a mild evidence cost (+1.65±0.29pp, three-seed mean), the Qwen3 scale with the largest positive RSFT test tax. Within the Qwen3 family at ≥8B, RSFT approaches evidence neutrality (−0.91±0.74pp at 8B, three-seed mean). Cross-architecture, LLaMA-3.1-8B shows partial compression: SFT tax of −4.94pp reduces to −1.46±0.40pp (two-seed mean) under RSFT, consistent with the cross-family variation observed in SFT evidence tax (Appendix F).

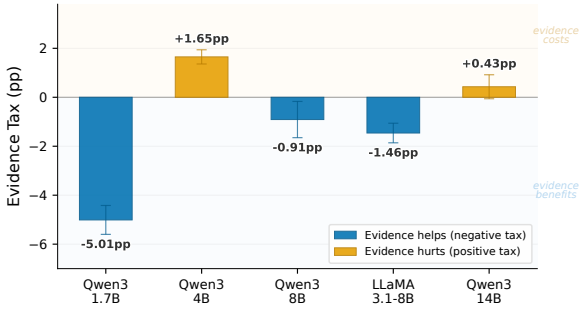


Figure 3: Inference-time evidence tax across model scales (RSFT models, Re-DocRED test). Negative values indicate evidence *benefits* extraction. Error bars show ± 1 std (all scales have ≥ 2 seeds).

4.5 Evidence Strategy Comparison

We compare joint generation against post-hoc BM25 retrieval (entity-constrained, k=2), using Evi-F1 (joint) with the DREEAM-protocol denominator (Ma et al., 2023). Post-hoc retrieval achieves higher Rel-F1 (48.76) by avoiding the evidence tax but lower Evi-F1 (joint) of 34.7 (EDCR 0.766). RSFT CED reaches 36.1 (EDCR 0.6487), a 1.4-point evidence advantage. CED reranking at inference (best-of-N=8 by CED score) achieves Rel-F1 48.64 and Evi-F1 81.88 (TP-only), approaching no-evidence extraction quality while preserving evidence.

4.6 Generalization

Cross-architecture validation on LLaMA-3.1-8B (Table 2) shows the largest RSFT gain (+5.45pp) with complete pairwise seed separation (§4.4). The RSFT improvement transfers across datasets: on Re-DocRED (Tan et al., 2022), RSFT achieves 41.45 Ign-F1 (+4.86pp over SFT 36.59), matching the DocRED gain (+5.08pp). Our best joint Evi-F1 (36.1, DREEAM-protocol) falls below supervised DREEAM (54.15% (Ma et al., 2023)), a gap driven by extraction performance rather than evidence quality (TP-only Evi-F1 exceeds 80%). On the Re-DocRED test set, RSFT Ign-F1 drops approximately 20pp relative to dev across all evaluated scales (e.g., 14B: 58.5→38.5), attributable to 2.82× higher annotation density in test (34.7 vs. 12.3 gold triples per document); model ranking is perfectly preserved (Spearman ρ=1.0; Table 9 in Appendix). Full results appear in Appendix H.

4.7 Discussion

EDCR (§3.5) validates as a diagnostic at both cross-method ($\rho = -0.96$, $n = 7$ unique methods, $p < 0.001$; Appendix L) and per-relation levels ($\rho = -0.79$, $n = 76$, $p < 0.001$; Appendix L), the latter orthogonal to Evi-F1 ($\rho = -0.06$); partial correlations controlling for shared components are virtually unchanged ($\rho = -0.9617$ vs. -0.9643). CED and flat-NLI produce comparable RSFT results (Rel-F1 45.63 vs. 45.61); stricter selection (k=1) outperforms k=3 by 1.02 Ign-F1 despite fewer examples (Appendix B). RSFT reduces truncation FN, hallucinated FP, and duplicates by 6–22% across scales, but evidence quality is unchanged ($\Delta < 0.4$pp; Appendix K): RSFT improves *what* the model extracts but not *how well* it grounds extractions—concrete evidence citation comparisons in Appendix M illustrate this pattern. Inference overhead analysis (Appendix N) reveals that evidence generation adds 16–24% wall-clock time at 4B–14B scale but paradoxically reduces inference time at 1.7B by 13.6%, as the evidence field constrains output structure and reduces prediction count by 1.75×.

5 Conclusion

We formalize evidence-grounded document-level relation extraction as a structured generation task and compare three post-training paradigms across three fully evaluated scales (1.7B, 4B, 8B) with 14B validation, two model families, and two datasets. The evidence tax is not a fixed penalty but a landscape shaped by capacity, optimization paradigm, and data quality filtering. RSFT improves mean extraction across all tested scales and reduces training variance at every evaluated configuration; a 1.7B RSFT model approaches 8B SFT performance at 4.7× fewer parameters. Future work should investigate targeted evidence rewards and extend evaluation to non-Wikipedia domains.

6 Limitations

We evaluate on two datasets (DocRED, Re-DocRED), both derived from Wikipedia; cross-domain evaluation is constrained by the absence of evidence-annotated DocRE datasets beyond Wikipedia, a limitation shared by all prior evidence-aware work (SAIS, DREEAM, EIDER). Evidence tax patterns span two model families (Qwen3, LLaMA) with single hyperparameter settings per RL method; other architectures, scales beyond 14B, or alternative KL coefficients may produce different decoupling behaviors. CED scoring operates at the document level using an external NLI model

(DeBERTa-v3); sensitivity to other NLI model families is unexplored, and sentence-level CED filtering disrupts co-reference chains (Evi-F1 −7.7pp).

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A Training Hyperparameters

Table 3 lists the hyperparameters for all training configurations. All methods use LoRA with rank r=64 and α=128 applied to all linear projections (attention and feed-forward). SFT and RSFT share the same optimizer settings; RSFT differs only in data construction (top-1 CED-selected outputs from N=8 candidates). GRPO and DPO use lower learning rates to stabilize post-SFT optimization.

B Full Ablation Results

Table 4 reports the complete RSFT ablation across scoring functions, top-k values, and random seeds. RSFT CED k=1 achieves a four-seed mean Ign-F1 of 44.66 (standard deviation 0.53), confirming robust improvement over the SFT s42 baseline (39.74). The three-seed mean reported in Table 1 (44.82, seeds 42/43/44) is slightly higher because it excludes the lowest seed (s45: 44.16). Random selection RSFT k=1 (41.14 Ign-F1), which samples from the same candidate pool without CED scoring, improves only +1.40pp over SFT s42 versus +4.92pp for CED RSFT, confirming that CED scoring precision drives the majority of the RSFT gain. Increasing k to 3 slightly degrades performance

Table 3: Training hyperparameters for all optimization paradigms. All experiments use Qwen3-4B as the base model with LoRA adaptation. A dash indicates that the parameter is not applicable to the given method.

Parameter	**SFT**	**RSFT**	**GRPO QLoRA**	**GRPO bf16**	**DPO**
Base model	Qwen3-4B	Qwen3-4B	Qwen3-4B	Qwen3-4B	Qwen3-4B
LoRA rank r	64	64	64	64	64
LoRA α	128	128	128	128	128
Learning rate	2e-5	2e-5	5e-6	5e-6	5e-7
Batch size (effective)	16	16	4	8	4
Epochs	3	3	–	–	1, 3*
Max steps	–	–	100	100	–
Gradient accumulation	4	4	16	4	4
Max new tokens	1024	1024	1024	1024	1024
Quantization	–	–	NF4 4-bit	bf16	bf16
Group size G	–	–	4	8	–
KL coefficient	–	–	0.01	0.01	–
β (DPO)	–	–	–	–	0.1
Candidates N	–	8	–	–	8
Top-k selection	–	1	–	–	–
Training samples	3,053	2,718	3,053	3,053	3,053
Scoring function	–	CED	CED	CED	CED

*DPO is trained for both 1 and 3 epochs to observe temporal dynamics of the evidence-extraction trade-off (Section 4.3).

(43.64 Ign-F1), consistent with the intuition that stricter selection yields higher-quality training data. The flat-NLI scoring variant produces comparable Rel-F1 (mean 45.61 vs. 45.63 for CED, a difference of 0.02pp). Both CED and flat-NLI improve substantially over the SFT baseline while maintaining or improving Evi-F1, confirming that the RSFT gains are driven by quality-based data selection rather than by the specific contrastive formulation.

NLI scoring comparison. Across all available seeds, CED RSFT achieves mean Rel-F1 of 45.63 (seeds 42, 43, 44, 45: range 45.14–46.21), and flat-NLI RSFT achieves mean Rel-F1 of 45.61 (seeds 42, 43: range 45.35–45.86). The 0.02pp difference suggests comparable performance: CED's contrastive design does not substantially outperform simpler NLI scoring for RSFT data selection. We adopt CED as the default because its contrastive formulation produces per-triple discrimination scores that support interpretable error analysis.

Top-k selection. Best-of-8 (k=1) yields mean Ign-F1 44.66 and Rel-F1 45.63, while top-3-of-8 (k=3) produces 43.64 and 44.54. The selected training data confirms the quality gap: k=1 data has mean extraction F1 of 61.57 and mean CED of 0.6683, versus 57.61 and 0.6205 for k=3. Despite k=3 providing 2.96× more training examples, the additional lower-ranked samples introduce noise that degrades final performance.

NLI scorer capacity. CED filtering is robust to NLI model capacity: replacing the default scorer (DeBERTa-v3-base, 86M parameters) with DeBERTa-v3-large (304M) yields only +1.10pp Ign-F1 (45.33 vs. 44.23, s42), a difference within the v3-base seed noise of 0.88pp (s42: 44.23, s44: 45.11). Other metrics confirm the pattern: Rel-F1 improves by +1.19pp (46.36 vs. 45.17), Evi-F1 by +0.20pp (81.94 vs. 81.74), while EDCR shifts by −0.39pp (0.6530 vs. 0.6569)—the inconsistent direction across metrics rules out systematic improvement from a larger scorer. This insensitivity supports v3-base as a cost-effective default, as a 3.5× larger scorer does not materially improve CED-based data selection.

Table 4: Full RSFT ablation on DocRED dev set (Qwen3-4B). CED and flat-NLI denote the scoring function used for candidate selection. Best Ign-F1 among evidence-generating methods in **bold**.

Configuration	**Ign-F1**	**Rel-F1**	**Evi-F1**	**EDCR**
SFT baseline (s42)	39.74	40.49	79.86	0.6982
SFT no-evidence	**47.21**	48.76	–	–
RSFT random k=1	41.14	41.98	79.64	0.6971
RSFT CED k=1 (s42)	44.23	45.17	81.74	0.6569
RSFT CED k=1 (s43)	45.12	46.01	80.63	0.6542
RSFT CED k=1 (s44)	45.11	46.21	80.50	0.6350
RSFT CED k=1 (s45)	44.16	45.14	81.74	0.6550
RSFT CED k=1 (mean)	44.66	45.63	81.15	0.6503
RSFT CED k=3	43.64	44.54	80.88	0.6635
RSFT flat-NLI (s42)	–	45.35	80.43	0.6431
RSFT flat-NLI (s43)	–	45.86	81.58	0.6505
RSFT flat-NLI (mean)	–	45.61	81.01	0.6468
RSFT CED k=1 v3-large (s42)	45.33	46.36	81.94	0.6530

Table 5: Detailed Rel-F1 and Evi-F1 across model scales on DocRED dev set. SFT and RSFT columns show multi-seed means where available (Qwen3 1.7B/4B/8B: 3-seed; 14B: two-seed mean; LLaMA SFT Rel-F1/Evi-F1: s42 only; LLaMA RSFT: 3-seed).

Model	SFT Rel-F1	SFT Evi-F1	RSFT Rel-F1	RSFT Evi-F1	RSFT EDCR
Qwen3-1.7B	45.86	82.85	52.81	82.63	0.6217
Qwen3-4B	44.39	79.61	45.80	80.96	0.6487
Qwen3-8B	51.87	82.94	56.07	81.54	0.6186
Qwen3-14B	56.11	82.92	59.79	83.28	0.5717
LLaMA-3.1-8B	55.14	81.51	57.75	82.84	0.5804

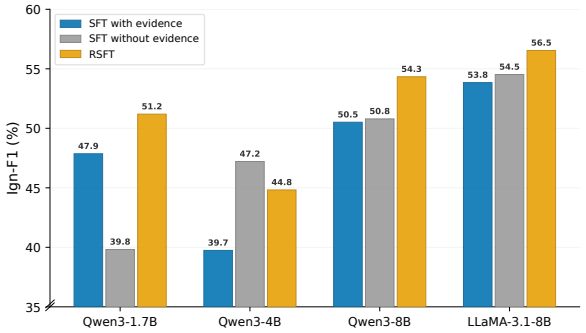


Figure 4: Evidence tax and RSFT gain across model scales. Bar heights show Ign-F1 for SFT with evidence (s42), SFT without evidence (s42), and RSFT (mean). The gap between no-evidence and SFT bars reverses direction between 1.7B and 4B, revealing the scale-dependent evidence tax. RSFT consistently narrows the gap toward the no-evidence upper bound.

C Scaling Details

Table 5 supplements Table 2 in the main text with per-scale Rel-F1 and Evi-F1 breakdowns. At 1.7B, RSFT improves Rel-F1 from 45.86 to 52.81 (+6.95, multi-seed mean) while achieving Evi-F1 82.63 and EDCR 0.6217. The 4B model shows the same pattern: Rel-F1 increases from 44.39 to 45.80 (+1.41, three-seed mean). For 8B, RSFT brings Rel-F1 from 51.87 to 56.07 (three-seed mean) with Evi-F1 of 81.54 and EDCR 0.6186, confirming tight coupling between extraction and evidence at this scale. At 14B, the SFT baseline achieves strong Rel-F1 (56.11) and Evi-F1 (82.92); RSFT improves Rel-F1 to 59.79 (two-seed mean) with Evi-F1 83.28 and EDCR 0.5717.

D Variance Compression Confidence Intervals

Table 6 reports 95% confidence intervals for SFT and RSFT standard deviations using the χ^2 distribution (N=3 seeds per configuration). At 1.7B, the confidence intervals are fully non-overlapping (SFT lower bound 1.45pp $>$ RSFT upper bound 0.54pp), providing the strongest evidence for variance compression. At the remaining three scales, intervals overlap but point estimates are consistently in the same direction (SFT std $>$ RSFT std at all four scales; sign test $p = 0.0625$, one-sided binomial, $n = 4$, underpowered). Cohen's d ranges from 0.46 (4B) to 3.31 (1.7B), with 1.7B and LLaMA-8B showing large effect sizes ($d > 2.5$).

Table 6: 95% confidence intervals for training variance (χ^2 method, N=3 seeds). Compression is the ratio of SFT to RSFT point estimates (unrounded).

Scale	SFT std	SFT 95% CI	RSFT std	RSFT 95% CI	Compr.	Cohen's d
Qwen3-1.7B	2.79	[1.45, 17.55]	0.09	[0.04, 0.54]	32.6×	3.31
Qwen3-4B	3.80	[1.98, 23.89]	0.51	[0.27, 3.21]	7.5×	0.46
Qwen3-8B	3.23	[1.68, 20.31]	0.50	[0.26, 3.15]	6.5×	1.79
LLaMA-3.1-8B	2.92	[1.52, 18.37]	0.64	[0.33, 4.01]	4.6×	2.58

Sign test: $p = 0.0625$ (one-sided binomial, $n = 4$; underpowered). The consistent direction (4/4 SFT $>$ RSFT), non-overlapping CIs at 1.7B, and large Cohen's d at 1.7B and LLaMA provide practical evidence for variance compression despite limited statistical power from N=3 seeds.

E Controlled Evaluation Subset

The 1.7B evidence tax comparison (§4.4) is confounded by differential truncation rates. At `max_new_tokens`=1024, the no-evidence model truncates 50.9% of outputs (501/985 documents) versus 31.9% (314/985) for the evidence model. To address this, we adopt a matched@2048 fair evaluation: both configurations are evaluated at `max_new_tokens`=2048, and the tax is computed on the intersection of documents where both produce format-compliant output. This yields 971/985 fair-subset documents, with a tax of −5.63pp (s42), confirming a genuine 1.7B evidence benefit. As additional validation, a stricter controlled subset at @1024 (excluding truncated or repetitive outputs; 442/985 documents) yields −6.2pp, with the stronger benefit attributable to the more aggressive filtering.

Table 7 provides a mechanistic breakdown across three scales (evaluated at @1024). Output duplication rate (fraction of documents containing ≥3 repeated triples or unique/total < 0.5) declines monotonically with scale for both training configurations, but the gap between no-evidence and with-evidence models narrows from 27.5pp at 1.7B to 3.7pp at 8B. This pattern supports the implicit regularization hypothesis: evidence generation forces smaller models to attend to supporting context, reducing degenerate repetition. At 4B, duplication is already moderate (31.1% with evidence), so the regularization benefit diminishes; under matched@2048 evaluation, the 4B training-time tax is approximately neutral (+0.25±0.47pp,

Table 7: Mechanistic breakdown of output quality across scales. Duplication rate: fraction of documents with degenerate outputs. Truncation Δ: no-evidence rate minus with-evidence rate (positive means no-evidence truncates more).

	Duplication Rate			Truncation	
Scale	**w/ evi**	**no-evi**	**Gap**	**Δ**	**Pattern**
1.7B	33.0	60.5	27.5	+19	no-evi worse
4B	31.1	43.2	12.1	+2	neutral
8B	29.2	32.9	3.7	−9	evi worse

three-seed mean). The truncation asymmetry reverses across scales: at 1.7B the no-evidence model truncates 19pp more often; at 8B the evidence model truncates 9pp more often due to the longer output format.

F Training-Time Evidence Tax

The evidence tax can be measured in two ways. The *training-time* tax compares separately trained models (SFT with evidence vs. SFT without evidence in the training output format). The *inference-time* tax (reported in Table 2) evaluates the same RSFT model with and without evidence at inference time.

Naïve comparisons at `max_new_tokens`=1024 are confounded by differential truncation rates (e.g., 50.9% vs. 31.9% at 1.7B). To control for this, we adopt a *matched@2048 fair evaluation*: both configurations are evaluated at `max_new_tokens`=2048, and the tax is computed on the intersection of documents where both models produce format-compliant output. Table 8 reports the matched@2048 training-time tax across Qwen3 scales.

The tax reveals a within-family capacity threshold: evidence generation benefits 1.7B (−5.63pp on 971/985 fair-subset documents) but is approximately neutral at ≥4B (+0.25±0.47pp at 4B across three seeds, +0.04pp at 8B, +0.03pp at 14B). At 1.7B, s42 is the strongest SFT seed (47.86 Ign-F1, +3.22pp above the three-seed mean of 44.64), so the benefit may be amplified relative to the population mean. LLaMA-3.1-8B SFT training-time tax is consistently negative but with high magnitude variance (−1.53 to −11.20pp, mean −5.19±5.25pp under matched@2048) and omitted from the table due to the s43 outlier (16.3% no-evidence format failure inflates its tax); cross-architecture patterns are addressed in §H.

Table 8: Training-time evidence tax under matched@2048 fair evaluation (DocRED dev Ign-F1). Both configurations evaluated at `max_new_tokens`=2048; tax computed on format-compliant intersection. Tax = no-evidence − with-evidence; positive = cost.

Model	**Fair w/ evi**	**Fair no-evi**	**Evi Tax**	**Subset**
Qwen3-1.7B	48.38	42.75	−5.63	971
Qwen3-4B	–	–	$+0.25 \pm 0.47^{(3)}$	≥982
Qwen3-8B	50.15	50.19	+0.04	984
Qwen3-14B	55.07	55.10	+0.03	985†

1.7B and 8B are s42; 4B reports three-seed mean ± std (per-seed range: −0.19 to +0.73pp); 14B evaluated at @1024 with 0% truncation for both configurations (†fair subset equals full dev set). Subset: documents in format-compliant intersection out of 985.

Evidence training and SFT seed variance. Evidence training's effect on SFT seed variance is also scale-dependent. At 4B, with-evidence SFT models show near-identical cross-seed spread (0.02pp) compared to no-evidence models (0.50pp), yielding 25× variance compression. At 8B, the compression is more modest (0.35pp vs. 0.48pp, 1.4×). However, at 1.7B the pattern reverses: with-evidence seed spread (0.63pp) exceeds no-evidence spread (0.20pp). This reversal likely reflects the extreme truncation rates at 1.7B (75.6% with-evidence, 94.4% no-evidence at `max_new_tokens`=1024): truncated outputs produce noisy extraction results whose variance depends on which triples appear before the cutoff point. The RSFT variance compression reported in §4.4 is measured relative to SFT with-evidence baselines and holds at all scales regardless of this SFT-internal pattern.

G Why RSFT Avoids Decoupling

The structural difference between selection and optimization explains why RSFT does not exhibit the decoupling observed in gradient-based methods. GRPO and DPO update model parameters through gradients that optimize a reward signal. When two sub-tasks (extraction and evidence) compete for gradient capacity, the optimizer can discover and exploit the trade-off between them, improving one at the expense of the other. RSFT operates differently: it selects complete outputs and trains on them with standard supervised loss. Because the selection is discrete (accept or reject a full response), there is no gradient pathway through which the model can learn to sacrifice extraction for evidence quality. Only outputs where both extraction and

evidence are jointly strong survive the CED filter. This property is not specific to RSFT's design but is a structural consequence of discrete selection over complete outputs: any selection-based method that filters on a joint quality criterion will inherit the same coupling guarantee. The implication for practitioners is direct: when optimizing for multiple coupled generation objectives, selection-based methods preserve inter-objective balance that gradient-based methods disrupt.

H Full Generalization Results

Cross-family transfer. LLaMA-3.1-8B confirms the scaling patterns observed with Qwen3. Its SFT baseline achieves a three-seed mean of 51.09 Ign-F1 (std 2.92pp; s42: 53.84, s43: 51.42, s44: 48.02). Under matched@2048 fair evaluation, the training-time tax is consistently negative across all three seeds (s42: -1.53, s43: -11.20, s44: -2.83; mean -5.19 ± 5.25pp), indicating that evidence generation consistently benefits LLaMA-3.1-8B. This contrasts with Qwen3-8B's approximately neutral tax (+0.04pp), suggesting that the evidence tax may be architecture-dependent at the 8B scale; however, the high magnitude variance (driven by s43, whose no-evidence configuration has a 16.3% format failure rate that inflates the tax estimate) precludes precise quantification. RSFT CED improves LLaMA to a three-seed mean of 56.54 Ign-F1 (std 0.64pp; s42: 57.14, s43: 56.62, s44: 55.87), a +5.45 point gain over SFT mean (51.09), with 4.6× variance compression (2.92pp → 0.64pp), consistent with the pattern at the 8B scale (Qwen3-8B: +4.14, 6.5× compression).

Cross-dataset robustness. On Re-DocRED (Tan et al., 2022), the SFT baseline achieves 36.59 Ign-F1 (Evi-F1 81.70, EDCR 0.6674). RSFT CED improves this to 41.45 Ign-F1 (Evi-F1 81.82, EDCR 0.6210), a +4.86pp gain that closely matches the +5.08pp improvement on DocRED (RSFT three-seed mean minus SFT s42). This consistency across datasets with different annotation protocols confirms that RSFT's benefit generalizes beyond training-set-specific artifacts. Evidence quality remains stable across datasets (Evi-F1 81.82 vs. 81.15 on DocRED), and the EDCR reduction (-0.046) mirrors the DocRED pattern, further supporting the robustness of CED-based selection.

Re-DocRED test set evaluation. Table 9 reports SFT and RSFT performance on the Re-DocRED test set. We observe a consistent performance gap between dev and test sets, with test Ign-F1 approximately 20 percentage points lower (RSFT: 1.7B 51.2→29.3, 4B 44.8→25.0, 8B 54.3→33.5, 14B 58.5→38.5, LLaMA-8B 56.5→35.5; SFT 1.7B: 44.6→25.8). This gap is attributable to annotation density: the test set contains 34.7 gold triples per document versus 12.3 in the dev set (a 2.82× ratio), imposing a harder recall requirement. The gap magnitude is stable across scales and architectures (range: 18.9–21.9pp), and model ranking is perfectly preserved (Spearman ρ=1.0), indicating that the absolute drop reflects the test annotation scheme rather than a model-specific deficiency. At 14B, the RSFT advantage observed on dev (+3.44pp) reverses on test ($\Delta=-0.37$pp); we note that the SFT model's 50.6% truncation rate implies longer outputs that generate more candidate triples to match the test set's dense annotation scheme.

Table 9: SFT and RSFT performance on Re-DocRED test set (Ign-F1 %). Dev Ign-F1 (RSFT) from Table 2. All test results use s42.

Model	Dev	SFT	RSFT	Δ	Tr. SFT	Tr. RSFT	Evi-F1
Qwen3-1.7B	51.19	25.79	29.33	+3.54	81.8	73.8	23.20
Qwen3-4B	44.82†	22.32	25.01	+2.69	64.6	42.8	18.38
Qwen3-8B	54.33	31.06	33.53	+2.47	82.0	1.8	25.26
Qwen3-14B	58.51	38.86	38.49	−0.37	50.6	13.4	26.88
LLaMA-3.1-8B	56.54	34.20	35.54	+1.34	65.4	60.0	24.57

†4B dev Ign-F1 is three-seed mean from original checkpoint; test results use retrained checkpoint (original unavailable for reproduction; retrained v2 achieves comparable dev performance, 43.88 vs. 44.23). Δ: RSFT − SFT. Evi-F1: RSFT joint (DREEAM protocol). Tr.: truncation rate (%). Dev is RSFT multi-seed mean from Table 2 (three-seed for 1.7B/4B/8B/LLaMA; two-seed for 14B).

Test-set evidence tax. Table 10 reports the evidence tax on the Re-DocRED test set, comparing SFT models trained with and without evidence in the output format. Test-set evaluation confirms the directional pattern observed on development data across all five model configurations: the tax is negative (beneficial) at 1.7B (-4.86pp) and LLaMA-8B (-4.94pp), and near-neutral at mid-scales (-0.13pp at 4B, -0.32pp at 14B). The 8B test tax (-1.71pp) shows slight variation from its near-neutral dev counterpart (+0.04pp), though both remain modest relative to the 1.7B and LLaMA benefits.

RSFT test-set evidence tax. Table 11 reports the inference-time evidence tax for RSFT models on the Re-DocRED test set (same RSFT model evaluated with vs. without evidence output; per-

Table 10: Evidence tax on Re-DocRED test set (Ign-F1 %). Tax = no-evi − with-evi; negative = benefit. All results use s42. Dev tax from Table 8.

Model	With-evi	No-evi	Test Tax	Dev Tax
Qwen3-1.7B	25.79	20.93	−4.86	−5.63
Qwen3-4B	22.32	22.19	−0.13	+0.25
Qwen3-8B	31.06	29.35	−1.71	+0.04
Qwen3-14B	38.86	38.54	−0.32	+0.03
LLaMA-3.1-8B	34.20	29.26	−4.94	−5.19

Table 11: RSFT inference-time evidence tax on Re-DocRED test set (Ign-F1 %). Tax = no-evi − with-evi; positive = cost. Per-seed s42 except where noted.

Model	RSFT w/ evi	RSFT no-evi	RSFT Tax
Qwen3-1.7B	29.02±0.45[(2)]	24.00±0.14[(2)]	−5.01±0.59[(2)]
Qwen3-4B	25.01	26.52	+1.51
Qwen3-8B	33.53	33.19	−0.34
Qwen3-14B	38.49	39.26	+0.77[†]
LLaMA-3.1-8B	35.54	33.80	−1.74

[(2)] Two-seed mean ± std (s42, s43). 1.7B no-evi truncation rate is 94.8% at s42 (vs. 73.8% with-evi), which may amplify the tax magnitude. [†] 14B shows a dev-test reversal (dev tax = −2.78±2.49pp, two-seed mean); the two-seed test mean (+0.43±0.49pp) is small relative to inter-seed variance (CI crosses zero), suggesting the direction signal is noise-dominated at this magnitude. All other scales preserve dev-test direction consistency.

seed s42 values). Figure 3 presents multi-seed means with error bars where ≥2 seeds are available. At 1.7B, RSFT preserves a strong evidence benefit (−5.01±0.59pp, two-seed mean), comparable to the SFT tax (−4.86pp). At 4B, the tax is mildly positive (+1.65±0.29pp, three-seed mean; s42: +1.51pp), the Qwen3 scale with the largest positive RSFT test tax; this contrasts with the near-neutral SFT tax at 4B (−0.13pp). At 8B, RSFT compresses the tax magnitude relative to SFT (−0.91±0.74pp, three-seed mean, vs. SFT −1.71pp). At 14B, the two-seed mean tax is +0.43±0.49pp (s42: +0.77pp, s43: +0.08pp), reversing sign relative to dev (−2.78±2.49pp, two-seed mean); the small test magnitude and high inter-seed variance may reflect noise or evaluation protocol differences given the 2.82× annotation density gap between dev and test. LLaMA-3.1-8B shows partial compression from SFT (−4.94pp → −1.46±0.40pp, two-seed mean).

Comparison to supervised methods. Table 12 directly compares encoder-based methods with our LLM-based approach. Encoder methods achieve substantially higher extraction performance (Rel-F1 63–65 vs. our 46–60), reflecting decades of architecture-specific optimization for the classification formulation. However, their evidence identification operates as binary sentence classification, producing Evi-F1 of 52–56 under a joint evaluation protocol. Our TP-only Evi-F1 exceeds 80%, indicating that when the model correctly identifies a triple, it produces higher-quality evidence than encoder methods. The joint Evi-F1 gap (36.1 vs. 54.15) is driven primarily by extraction performance rather than evidence quality: false-positive and false-negative triples dilute the denominator. The extraction gap varies with scale (16–18pp Ign-F1 at 4B, narrowing to <5pp at 14B), reflecting the rapid scaling of generative approaches relative to encoder methods. Further closing this gap at smaller scales requires improving extraction capability of LLM-based methods, an orthogonal direction to the evidence faithfulness problem we address.

Table 12: Comparison with encoder-based methods on DocRED dev set. Encoder methods use RoBERTa-large; LLM methods use Qwen3. Evi-F1 for encoder methods uses joint evaluation (all triples in denominator); our joint Evi-F1 follows the same protocol for fair comparison. TP-only Evi-F1 conditions on correctly predicted triples. Rel-F1 is reported in percentage for all methods.

Method	Type	Ign-F1	Rel-F1	Evi-F1 (joint)	Evi-F1 (TP)
ATLOP	Encoder	61.32	63.18	–	–
EIDER	Encoder	62.34	64.27	52.54	–
SAIS	Encoder	62.23	65.17	55.84	–
DREEAM	Encoder	61.71	63.49	54.15	–
Ours (4B RSFT)	LLM	44.82	45.80	36.1	80.96
Ours (8B RSFT)	LLM	54.33	56.07	–	81.54
Ours (14B RSFT)	LLM	58.51	59.79	48.57	83.28

Table 13: Evidence-extraction decoupling across RL paradigms (Qwen3-4B). Severity indicates the degree of extraction collapse relative to evidence improvement.

Paradigm	Config	Rel-F1	Evi-F1	EDCR	Severity
GRPO	QLoRA G=4	1.38	85.04	0.9876	Catastrophic
GRPO	bf16 G=8	45.49	80.86	0.6318	Mild
DPO	bf16 1ep	33.37	80.90	0.8123	Gradual
DPO	bf16 3ep	31.45	84.84	0.7503	Gradual
SFT	baseline[‡]	44.39	79.61	0.6797	None
RSFT	CED[‡]	45.80	80.96	0.6487	None

[‡] Three-seed mean. DPO rows evaluated at `max_new_tokens`=2048 (matched protocol); all other rows at 1024. Severity reflects training trajectory stability (propensity for extraction collapse during optimization), not the final-checkpoint EDCR value.

I Decoupling Spectrum Data

J GRPO Training Dynamics

K Error Analysis: SFT vs. RSFT

Table 14 decomposes the extraction improvements from RSFT into error categories across three scales. Three mechanisms drive the gains consistently:

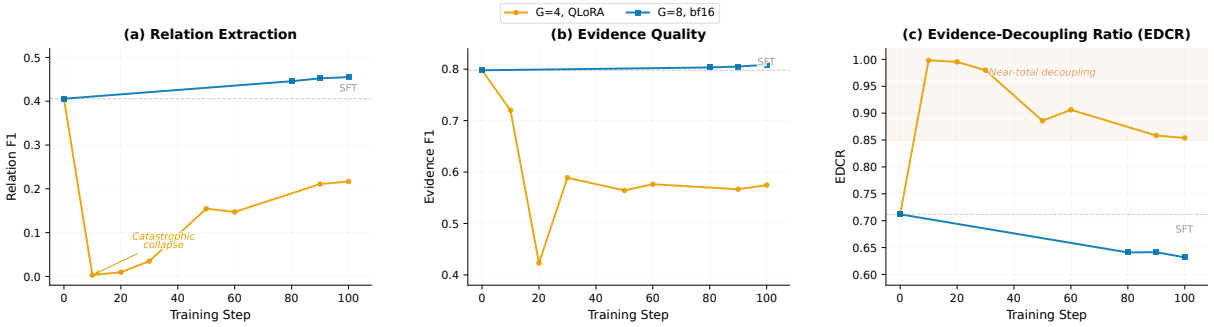


Figure 5: Training dynamics of GRPO under different configurations. (a) Relation F1 across checkpoints: G=4 QLoRA suffers catastrophic collapse at step 10, while G=8 bf16 remains stable. (b) Evidence F1 stays intact for G=8 bf16, confirming the generation mechanism itself does not fail. (c) Evidence-Decoupling Ratio (EDCR) reveals near-total decoupling in G=4 QLoRA (extraction collapses to Rel-F1 1.4), while G=8 bf16 maintains healthy ratios.

Truncation false negatives. SFT models frequently hit the generation length limit, producing incomplete outputs that miss triples. RSFT training on compact, high-quality outputs reduces this truncation rate. The effect is strongest at 1.7B (-841 truncation FN, -22%), where the SFT model's output length is least controlled.

Hallucinated false positives. CED-based filtering removes training examples with unfaithful predictions, reducing the rate of hallucinated triples at inference. LLaMA benefits most (-593 hallucinated FP, -11.8%), suggesting cross-architecture variation in hallucination susceptibility.

Duplicate predictions. RSFT training data, selected for quality, contains fewer repetitive patterns. This propagates to inference, reducing duplicate triples by 11–20% across scales, with 1.7B showing the largest reduction (-20.2%).

Evidence quality unchanged. Despite substantial extraction improvements, evidence exact-match F1 changes by less than 0.4pp at every scale. RSFT selects for joint extraction-evidence quality, but the evidence signal in CED scoring does not translate into measurable evidence improvement. This suggests that evidence grounding requires targeted optimization (e.g., evidence-specific rewards in RL) rather than data selection alone.

Residual difficulty. The proportion of "other" false negatives (triples missed for reasons beyond truncation or formatting) increases by 13–15% in relative terms after RSFT. As easy errors (truncation, duplication) are eliminated, the remaining error distribution shifts toward harder cases: cross-sentence reasoning, implicit relations, and long-tail entity pairs. The relation types with the largest RSFT gains (participant, conflict, league, screenwriter, operator, contains_administrative_territory) involve cross-sentence reasoning or long-tail entities, consistent with RSFT's selection pressure toward diverse, high-quality outputs.

Table 14: Error pattern changes from SFT to RSFT across scales. $\Delta\%$ shows relative change. Truncation FN, hallucinated FP, and duplicates all decrease; evidence quality and residual FN difficulty are unchanged.

Error Category	Direction	1.7B	4B	8B / LLaMA
Truncation FN	$\downarrow$	-22%	-16%	-18%
Hallucinated FP	$\downarrow$	-8%	-6%	-12%
Duplicates	$\downarrow$	-20%	-11%	-14%
Evidence (exact)	$\approx$	$<$0.4pp	$<$0.4pp	$<$0.4pp
FN: other	$\uparrow$	+13%	+15%	+14%

L Per-Relation EDCR Distribution

Computing EDCR for each of the 76 relation types with $\geq$5 evaluation samples (from the RSFT CED s42 model) reveals that per-relation EDCR correlates strongly with per-relation Rel-F1 ($\rho = -0.79$, $n = 76$, $p < 0.001$), confirming its utility as a difficulty indicator. Filtering to types with $\geq$5 predictions strengthens the correlation ($\rho = -0.85$, $n = 74$, $p < 0.001$). Per-relation EDCR is statistically independent of Evi-F1 ($\rho = -0.06$, $p = 0.58$), and Evi-F1 itself shows no association with Rel-F1 ($\rho = -0.01$, $p = 0.91$). These null correlations establish that evidence quality and extraction ability are fully decoupled across relation types. Extreme cases illustrate the diagnostic power: `date_of_birth` has EDCR $= 0.118$ (highly faithful, with extraction and evidence tightly coupled), while `head_of_government` reaches 0.956 (highly decoupled). This per-relation analysis avoids the shared-component concern present in cross-method

EDCR-vs-Rel-F1 correlation, providing independent validation of EDCR's diagnostic utility. A permutation test (10,000 iterations) on the per-relation correlation yields $p < 0.0001$, confirming that the association is not an artifact of the shared TP component.

Partial correlations controlling for TP. Because EDCR shares TP (true positives) in both numerator and denominator with Rel-F1, we report partial correlations controlling for Evi-Recall$_{\text{TP}}$. At the cross-method level ($n = 17$ seed-specific configurations), the partial Spearman $\rho = -0.8541$ ($p = 0.000025$), virtually unchanged from the raw $\rho = -0.8554$ (0.15% reduction). Removing the GRPO QLoRA outlier yields partial $\rho = -0.8681$ ($n = 16$, $p = 0.000027$). Aggregating to $n = 7$ unique methods: raw $\rho = -0.9643$, partial $\rho = -0.9617$. These results confirm that the EDCR–Rel-F1 association is not driven by the shared TP component. Note that seed-level configurations ($n = 17$) are not fully independent; the $n = 7$ unique-method analysis provides a conservative estimate.

Similar-TP, different-EDCR pairs. To further validate that EDCR captures information beyond what TP alone provides, we identify configuration pairs with similar Evi-Recall$_{\text{TP}}$ ($|\Delta| < 5$pp) but substantially different EDCR ($|\Delta| > 5$pp). Forty-eight such pairs exist across all configurations. The most illustrative: DPO-CED 1-epoch (Evi-Recall$_{\text{TP}} = 0.833$) vs. RSFT CED LLaMA s43 (Evi-Recall$_{\text{TP}} = 0.829$) differ by only 2.4pp in TP-conditioned evidence recall but 13.6pp in EDCR, reflecting the substantial extraction gap between these methods. This pattern demonstrates that EDCR captures extraction-evidence alignment that aggregate TP metrics miss.

M Qualitative Analysis

We present representative examples illustrating three phenomena: evidence quality differences between SFT and RSFT, GRPO format gaming under catastrophic collapse, and per-relation EDCR diagnostics.

Case 1: Evidence quality. Table 15 compares evidence citations from SFT and RSFT on three documents. SFT tends to cite many sentences indiscriminately, including distractors unrelated to the target triple, while RSFT selects precisely the gold evidence sentences. For *America's Sweetheart*, SFT cites 6 sentences (precision 0.17) whereas RSFT cites only the single correct sentence (precision 1.00). The pattern is consistent: SFT achieves evidence precision of 0.00–0.20 across these examples, while RSFT achieves 1.00 in all three cases.

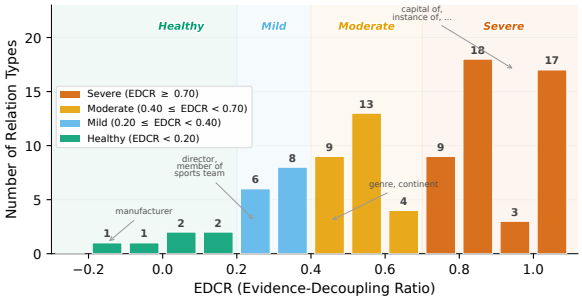


Figure 6: Per-relation EDCR distribution across all 93 relation types observed in the development set (of 96 defined in DocRED). The correlation analysis in the text uses the 76 types with ≥5 samples for statistical reliability. Red bars indicate EDCR ≥ 0.70 (severe decoupling), green bars indicate EDCR < 0.30 (healthy evidence grounding). Seven types exhibit EDCR = 1.0.

Table 15: Evidence citation comparison between SFT and RSFT. Precision measures the fraction of cited sentences that match gold evidence.

Document	Triple	SFT Prec.	RSFT Prec.	SFT #Cited
Oleg Tinkov	(Oleg Tinkov, country of citizenship, Russia)	0.00	1.00	3
America's Sweetheart	(record label, country, United States)	0.17	1.00	6
Jade (Mortal Kombat)	(Jade, present in work, Mortal Kombat)	0.20	1.00	5

Case 2: GRPO format gaming. Under catastrophic collapse (GRPO QLoRA G=4), the model abandons the DocRED 96-type relation ontology and generates free-text relation labels. On the document *Something to Believe In*, the model produces 35 predicted triples with relations such as "is a", "honor", and "location", none of which belong to the target ontology. All 35 predictions are incorrect. Similarly, for *Józef Piłsudski Monument*, 28 predictions use non-standard relations with 0 correct matches. The model retains the structural format of triple-with-evidence output but fills it with fabricated relation types. The model produces well-formed evidence for these non-existent relations.

Case 3: EDCR diagnostics. Per-relation EDCR identifies specific relation types where extraction and evidence are misaligned. For `date_of_birth` (EDCR = 0.118), the model's evidence consistently matches gold annotations: both extraction and evidence point to the same biographical sentence. In contrast, `notable_work` (EDCR = 0.941) exhibits severe decoupling: the model cites sentence [0] (a general introduction) instead of

Table 16: Inference time (seconds) for SFT with vs. without evidence output. Overhead = (with-evi − no-evi) / no-evi; negative means with-evidence is faster. 1.7B–8B use batch size 4; 14B uses batch size 1.

Model	With-evi	No-evi	Overhead	#Pred (w/no)
Qwen3-1.7B	6,424	7,438	−13.6%	25.5k / 44.6k
Qwen3-4B	8,887	7,606	+16.8%	24.7k / 25.6k
Qwen3-8B	10,486	10,901	−3.8%	25.0k / 31.1k
Qwen3-14B	18,011	14,498	+24.2%	—

the gold evidence sentence [3] (which contains the actual work attribution). High-EDCR relations tend to involve indirect or multi-hop reasoning where surface-level context misleads the evidence generator.

N Inference Efficiency Analysis

Table 16 reports wall-clock inference time for SFT models with and without evidence output under matched evaluation settings (batch size 4, `max_new_tokens`=2048, DocRED dev, 985 documents, single RTX 5090 GPU).

The evidence overhead is scale-dependent and non-monotonic. At 1.7B, removing evidence *increases* inference time by 13.6%: without the structural constraint of evidence fields, the model generates 1.75× more predictions (44.6k vs. 25.5k), producing verbose output that takes longer to decode. At 14B, evidence generation adds 24.2% overhead, consistent with the expected cost of generating additional tokens. The crossover occurs around 4B–8B, where the prediction count differential narrows and the costs approximately balance.

Inference time scales sub-linearly with model size: 1.7B→8B (4.7× parameters) increases wall-clock time by only 1.63× at batch size 4, reflecting the memory-bound nature of autoregressive generation. Test-set inference is 2–3× slower per document than dev (e.g., 24–42 s/doc test vs. 7–18 s/doc dev at batch size 1), attributable to the 2.82× higher annotation density in Re-DocRED test requiring longer generated outputs.

O Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform the related-work section.